Career Launcher MBA

QA Numbers-02

Factors

Question 1

If ‘N’ is a natural number, then how many values of ‘N’ are possible such that

$\frac{(15N^2 + 13N + 36)}{N}$ is also a natural number?

(1) 8

(2) 9

(3) 10

(4) 6

Question 2

What is the largest positive integer n such that $\frac{n^2+7n+12}{n^2-n-12}$ is also a positive integer?

(1) 16

(2) 12

(3) 6

(4) 8

Question 3

The number of positive integers n in the range $12 \leq n \leq 40$ such that the product $(n - 1)(n - 2)\ldots3..2.1$ is not divisible by n is

(1) 5

(2) 7

(3) 13

(4) 14

Question 4

Consider $N = 2^7 \times 3^6 \times 5^3$

a. Find the total number of factors of ‘N’.
b. Find the total number of factors of ‘N’ which are odd.
c. Find the total number of factors of ‘N’ which are even.
d. Find the total number of factors of ‘N’ which are composite.
e. Find the total number of factors of ‘N’ which are multiple of 3.
f. Find the total number of factors of ‘N’ which are multiple of 12.
g. Find the total number of factors of ‘N’ which are perfect square.
h. Find the total number of factors of ‘N’ which are perfect cube.
i. Find the total number of factors of ‘N’ which has unit’s digit zero.
j. Find the total number of factors of ‘N’ whose last two digits are zero.
k. Find the sum of all factors of ‘N’.
l. Find the product of all factors of ‘N’.

Question 5

Consider $N = 2^2 \times 6^4 \times 5^6 = A \times B$

a. In how many ways ‘N’ can be written as a product of two numbers?

b. In how many ways ‘N’ can be written as a product of two different numbers?

c. In how many ways ‘N’ can be written as a product of two numbers which are odd?

d. In how many ways ‘N’ can be written as a product of two numbers which are even?

e. In how many ways ‘N’ can be written as a product of two numbers such that one of them is even and the other is odd?

f. In how many ways ‘N’ can be written as a product of two numbers which are co-prime to each other?

Question 6

A is a factor of 720. A itself has exactly three factors. How many values of A are possible?

Question 7

A number has 72 factors. What can be the maximum and minimum number of prime factors of this number?

(1) 71 & 1

(2) 6 & 2

(3) 5 & 2

(4) None of these

Question 8

How many factors of $2^8 \times 3^6 \times 5^4 \times 10^5$ are multiples of 120?

(1) 594

(2) 660

(3) 540

(4) None of the mentioned options are correct

Question 9

How many factors of $2^4 \times 3^5 \times 10^4$ are perfect squares which are greater than 1?

Question 10

A positive number is such that it has a total of four factors and the sum of the factors, excluding the number itself, is 31. What is the sum of all such numbers?

(1) 125

(2) 591

(3) 716

(4) 898

Question 11

Find the number of divisors of $N = 2^7 \times 3^5 \times 5^3$ which are of the form $4t + 1$, where t is a natural number.

(1) 6

(2) 12

(3) 11

(4) None of these

Question 12

A number ‘n’ has nine factors below its square root. ‘N’ has how many factors?

(1) 17

(2) 18

(3) 19

(4) 18 or 19

Question 13

A perfect square number ‘N’ has nine factors below its square root. Find the number of factors of ‘N^5’.

Question 14

In how many ways 1000 can be written as a sum of two or more consecutive natural numbers?

Question 15

Consider the set S = {1, 2, 3, . . . , 100}. How many arithmetic progressions can be formed from the elements of S that start with 1 and end with 100 and have at least 3 elements?

(1) 9

(2) 8

(3) 7

(4) 5

Question 16

The number of integers x such that $0.25 \leq 2^x \leq 200$, and $2^x + 2$ is perfectly divisible by either 3 or 4, is

Question 17

For all possible integers n satisfying $2.25 \leq 2 + 2^{n+2} \leq 202$, the number of integer values of $3 + 3^{n+1}$ is

Question 18

How many pairs (a, b) of positive integers are there such that $a \leq b$ and $ab = 4^{2017}$?

(1) 2017

(2) 2018

(3) 2019

(4) 2020

Question 19

Let A be the largest positive integer that divides all the numbers of the form $3^k + 4^k + 5^k$, and B be the largest positive integer that divides all the numbers of the form $4^k + 3(4^k) + 4^{k+2}$, where k is any positive integer. Then (A + B) equals

Question 20

A school has less than 5000 students and if the students are divided equally into teams of either 9 or 10 or 12 or 25 each, exactly 4 are always left out. However, if they are divided into teams of 11 each, no one is left out. The maximum number of teams of 12 each that can be formed out of the students in the school is

Thanks!